AS400 (IBM i) Notes: Objects, Libraries, Jobs, Commands, Syntax & Function Keys

1. AS400 Objects

What is an Object?
Everything in AS400 is stored as an object. Objects contain data or programs.

Common Object Types:
* *FILE – Database file
* *PGM – Program
* *LIB – Library
* *JOBQ – Job Queue
* *OUTQ – Output Queue
* *DTAARA – Data Area
* *USRPRF – User Profile

Commands:
WRKOBJ OBJ(*ALL/*ALL) – Work with objects
DSPOBJD OBJ(MYLIB/*ALL) OBJTYPE(*FILE) – Display object description
DLTOBJ OBJ(MYLIB/TESTFILE) OBJTYPE(*FILE) – Delete object

Example:
DSPOBJD OBJ(QGPL/CUSTFILE) OBJTYPE(*FILE)

2. Libraries

What is a Library?
A library is like a folder that stores AS400 objects.

Commands:
CRTLIB LIB(TESTLIB) – Create library
DSPLIB LIB(TESTLIB) – Display library
WRKLIB LIB(TESTLIB) – Work with library
DLTLIB LIB(TESTLIB) – Delete library

Library List Commands:
DSPLIBL – Display library list
ADDLIBLE LIB(TESTLIB) – Add library to library list
RMVLIBLE LIB(TESTLIB) – Remove library

Example:
CRTLIB LIB(STUDENTLIB) ADDLIBLE LIB(STUDENTLIB)

3. Jobs

What is a Job?
A job is a unit of work performed by the system.

Types of Jobs:
Interactive Job
Batch Job
Autostart Job
Prestart Job

Commands:
WRKACTJOB – Active jobs
WRKJOB – Work with current job
DSPJOB – Display job information
SBMJOB – Submit batch job
ENDJOB – End job

Syntax:
SBMJOB CMD(CALL PGM(MYLIB/MYPGM)) JOB(TESTJOB)

Example:
SBMJOB CMD(CALL PGM(QGPL/CUSTPGM)) JOB(CUSTJOB)

4. Job Queues

Job Queue Commands:
CRTJOBQ JOBQ(MYLIB/MYJOBQ)
WRKJOBQ JOBQ(MYLIB/MYJOBQ)
DSPJOBQ JOBQ(MYLIB/MYJOBQ)

Submit Job to Queue:
SBMJOB CMD(CALL PGM(MYLIB/TESTPGM)) JOBQ(MYLIB/MYJOBQ)

5. Frequently Used Commands

DSPLIBL – Display library list
WRKLIBL – Work with library list
WRKOBJ – Work with objects
WRKMBRPDM – Work with members
STRSEU – Source Entry Utility
CRTBNDRPG – Compile RPGLE program
CRTPF – Create physical file
CRTLF – Create logical file
DSPFD – Display file description
DSPPFM – Display physical file member

6. Important Function Keys

F3 – Exit
F4 – Prompt command parameters
F5 – Refresh
F6 – Create/Add
F9 – Retrieve previous command
F10 – Additional parameters/details
F11 – Display more information / switch views
F12 – Cancel/Return
F16 – Repeat position
Enter – Execute command

Practical Usage:
Type CRTLIB and press F4 to see all parameters.
Press F9 to retrieve previous command.
Press F3 to exit current screen.

7. Real-Time Production Support Examples

1. Check active jobs: WRKACTJOB

2. Verify library list: DSPLIBL

3. Check file details: DSPFD FILE(MYLIB/CUSTFILE)

4. Submit batch processing: SBMJOB CMD(CALL PGM(MYLIB/NIGHTPGM))

5. Review current job logs: DSPJOBLOG

8. Interview Questions

What is an object in AS400?
Difference between library and object?
Difference between interactive and batch jobs?
Purpose of SBMJOB?
What is a job queue?
What does DSPLIBL do?
Why do we use F4 and F9?